Writing & Interpreting Linear Models

WARM UPS

Name: _____ Date: _____

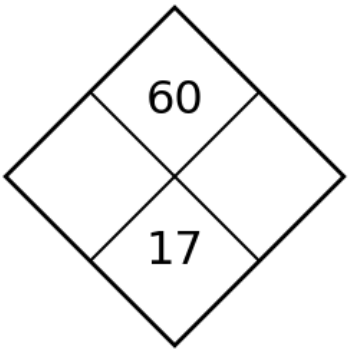
1 | Solve

Solve for x. Show each step, then check your answer.

$$2(3x - 4) + 5 = 27$$

2 | Diamond

Complete the diamond.



Writing & Interpreting Linear Models

WARM UPS

3 | Spiral Review

Calculate slope between (1,7) and (3,15).

4 | Spiral Review

Is {(0,2),(1,4),(2,6)} a function? List domain and range.

Writing & Interpreting Linear Models

LEARNING FOCUS & NOTES

Learning focus | LG 03-ican-2 / LG 03-ican-5

- I can write y = mx + b from a description, table, or two points.
- I can explain what the slope and starting value mean.
- I can check whether a model makes sense for a stated input.

Vocabulary | Highlight the important words

Linear model

An equation, table, or graph using constant change to represent a situation.

Slope-intercept form

The equation y = mx + b, where m is slope and b is the output at x = 0.

Slope / rate m

The change in output for each 1-unit input increase; include units.

y-intercept / starting value b

The value of y when x = 0; often the starting amount in context.

Variable

A letter representing a changing or unknown quantity, with an assigned meaning.

Domain restriction

A limit on which inputs make sense, based on the situation or mathematical rule.

Meaning & Method | Rewritten review notes

1. Read the story first: choose an input x and an output y with meaningful units.

2. Identify the starting amount b at input zero, then identify the constant change m per one input unit. Write y = mx + b.

3. Check the model using an input and output from the context. State restrictions such as nonnegative whole-number counts.

These are original, concise review notes aligned to the course lessons; they are not a reproduction of the textbook pages.

Writing & Interpreting Linear Models

EXAMPLES & YOU TRY ITS

Example

A printing service charges a $12 setup fee plus $3 per poster. Write the cost equation, explain the fee and per-poster rate, and find the cost for 5 posters.

Posters p	Cost C ($)
0	12
1	15
2	18

Write the cost equation. Explain the fees and find the cost of 5 posters.

You Try It

A studio charges a $9 reservation fee plus $4 per hour. Complete the table, write the cost equation, explain the slope and starting value, and calculate the cost for 3 hours.

Hours h	Cost C ($)
0	___
1	___
2	___

Write the equation, interpret the $9 and $4, and find cost for 3 hours.

How do the setup fee and rate appear in your equation?

Writing & Interpreting Linear Models

LAUNCH

Which rental plan fits?

Two bike rental plans are offered. Plan A costs $6 plus $4 per hour. Plan B costs $10 plus $3 per hour. Compare their costs at the times shown in the table.

Hours h	Plan A ($)	Plan B ($)
0	___	___
1	___	___
2	___	___
3	___	___

Plan A: $A = 4h + 6$

Plan B: $B = 3h + 10$

Use the table to make your comparison for these specific times.

1 | What does each plan's slope mean? What does each starting value mean?

2 | Which is cheaper at 1 hour? 3 hours? Explain using numbers and words.

Partner share: Compare reasoning, not just answers. What would convince someone else your conclusion is correct?

Writing & Interpreting Linear Models

REVIEW & PREVIEW

1 ★ A membership costs $5 per month plus $2 for each movie rented. Write a model for total monthly cost C for n movies. Interpret m and b.

2 The table has points (0,15), (1,13), (2,11), (3,9). Write y = mx + b and interpret its trend.

3 ★ Write the equation of a line through (0,4) and (2,10). Show work for m and b.

Writing & Interpreting Linear Models

REVIEW & PREVIEW

4 Choose which rule matches outputs 4, 7, 10 at inputs 0, 1, 2: (a) y=3x+4 (b) y=4x+3 (c) y=7x+4. Explain.

5 Position is modeled by p = 20 - 5t (meters, seconds). Interpret 20 and -5; predict position at t = 3.

6 Spiral + model check: Is {(0,7),(1,9),(2,11),(2,11)} a function? What linear rule fits these pairs?

Evidence check: Star a question you can explain confidently. Circle one you want to revisit on Monday, November 9.

Writing & Interpreting Linear Models

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Writing & Interpreting Linear Models

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Writing & Interpreting Linear Models

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Writing & Interpreting Linear Models

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ALGEBRA 1 | PROJECTABLE QUESTION

Q5 | DAY 4

Writing & Interpreting Linear Models

Position is modeled by p = 20 - 5t (meters, seconds). Interpret 20 and -5; predict position at t = 3.

5 / 6 • Same question as Review & Preview

ALGEBRA 1 | PROJECTABLE QUESTION

Q6 | DAY 4

Writing & Interpreting Linear Models

Spiral + model check: Is {(0,7),(1,9),(2,11),(2,11)} a function? What linear rule fits these pairs?

6 / 6 • Same question as Review & Preview